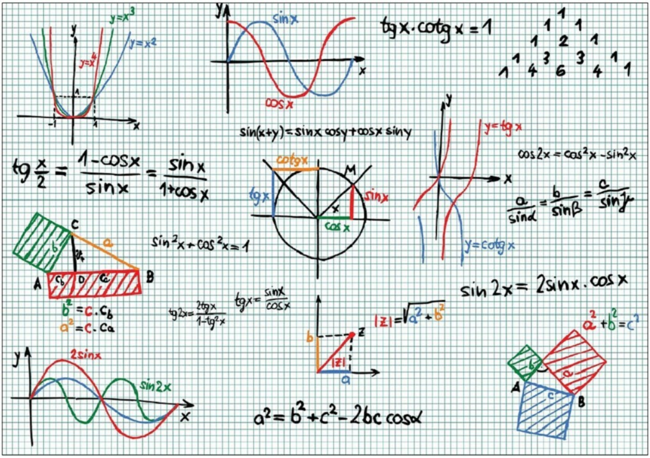
Use Maxima for the Simplification of Expressions

As covered in an earlier article, Maxima is a computer algebra system based on Macsyma and written in Lisp. In this article, the 15th in the series on using open source in mathematics, the author demonstrates the use of Maxima in the simplification of expressions.



Expression simplification can be done in a variety of ways. Let's start with the simple methods and then move on to more powerful ones.

Real number simplification

An example of a basic simplification is to convert a given number into a rational number, using *ratsimp()*, as follows:

```
$ maxima -q
(%i1) ratsimp(9);
(%o1)                              9
(%i2) ratsimp(10.0);
rat: replaced 10.0 by 10/1 = 10.0
(%o2)                             10
(%i3) string(ratsimp(4.2)); /* string to print on a line */
rat: replaced 4.2 by 21/5 = 4.2
(%o3)                             21/5
(%i4) string(factor(3.2));
rat: replaced 3.2 by 16/5 = 3.2
(%o4)                              2^4/5
(%i5) string(ratsimp(4.3333));
rat: replaced 4.3333 by 43333/10000 = 4.3333
(%o5)                          43333/10000
(%i6) string(ratsimp(4.3333333));
rat: replaced 4.3333333 by 13/3 = 4.333333333333333
(%o6)                             13/3
(%i7) quit();
```

Another example would be to check whether a number is an integer using *askinteger()*. And if it is, find out if it is